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- Module 3 Test
- Reading: Instructions for test and data to be used when answering questions 10 min
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- Course Conclusion

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Module 3 Test questions and your answers

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Due Sep 29, 11:59 PM IST Attempts 2 every 3 hours

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Your grade 88%

1. Choose all the correct answers below.

12 / 12 points

☐

The principal components transform is a good feature selection techniques when the data classes tend to be distributed along the main diagonal of the feature space.

☒

See figure 1 in the instruction sheet. Block (a) indicates that there is high correlation among the first 35 bands of a hyperspectral sensor. Block (b) indicates that there is strong water absorption between bands 35 and 72.

☒

Correct

That is correct

☒

See figure 1 in the instruction sheet. Block (b) indicates that there is strong water absorption between bands 35 and 72.

☒

This should not be selected

That is incorrect. That block indicates that bands 36-72 have little correlation with bands 1-35.

☐

Feature reduction is an important step in remote sensing image classification because it means we can then use any classification algorithm and expect good results.

2. Examine the error matrix in Figure 2 of the instruction sheet and choose the correct answers below.

6 / 6 points

☒

The average user's accuracy is 77.3%

☒

Correct

Yes that is correct.

☐

The average producer's accuracy is the same as the average user's accuracy.

☐

The overall classification accuracy is given by averaging the row and column sums.

☒

The class prior probabilities can be calculated from the column sums.

☒

Correct

Yes that is correct. The column sums are the totals from each ground cover class; we assume that the selected reference data reflects the actual scene proportions.

☒

The overall classification accuracy is 77.3%

☒

Correct

That is correct. It is the sum of the diagonal entries divided by 441 for this example.

3. Choose the correct answers below.

9 / 9 points

☒

The forward velocity of an imaging radar platform has to be synchronised with the rate at which the ranging pulses are transmitted so that continuous coverage of the terrain can be achieved.

☒

Correct

That is correct

☐

The look angle is the same as the incidence angle in an imaging radar system.

☐

The radar cross section of a resolution element is the same as its scattering coefficient.

☐

Speckle in a radar image is caused by the presence of large dominant scatterers in all resolution elements.

☒

The duration of the ranging pulse defines the range resolution of an imaging radar system

☒

Correct

That is correct

4. See figure 3 in the instruction sheet and choose the correct answers below

6 / 9 points

☐

The two lines of white dots in the bottom image just above the white arrow are satellites passing under the radar platform.

☒

The two lines of white dots in the bottom image just above the white arrow are oil platforms behaving as dihedral corner reflectors

☒

Correct

That is correct.

☐

The two lines of white dots in the bottom image just above the white arrow are a two moving ships being detected by successive sets of ranging pulses

☒

Sea surface features are more easily seen in the top image because the smaller incidence angle gives a smaller wave number in the sea surface spectrum.

☒

Correct

That is correct

☐

Sea surface features are more easily seen in the top image because on the day of acquisition there were strong winds.

☐

Terrain relief is less evident in the bottom image because of the radar illumination direction.

☒

Ignoring the relief distortion, the shading visible in the land surface in the bottom image is the result of the stronger dependence of surface scattering on surface roughness at larger incidence angles

☒

Correct

That is correct.

5. Choose the correct answers below.

8 / 8 points

☐

Shadows are not a problem for bistatic radar.

☐

Radar tomography allows topographic detail to be tracked in time.

☒

Radar tomography requires some penetration into the medium being imaged

☒

Correct

That is correct. Tomography reveals vertical structure so some penetration is needed.

☒

The altitude of the radar platform is important in topographic change detection in along-track interferometry.

☒

This should not be selected

That is incorrect. The absolute altitude does not affect the phase change between acquisitions caused by changes in the slant range.

☐

Relief distortion occurs in both ground range and slant range imagery

☒

A good general purpose look angle to use in an imaging radar system for land studies would be about 35 degrees.

☒

Correct

Yes that is correct. At small angles there is less discrimination of surface roughness variations and terrain distortion becomes a significant problem, whereas if it is too large shadowing becomes a problem.

6. Examine figure 4 in the instruction sheet and choose the correct answers below

8 / 8 points

☐

The multiple images of the bridge are the result of several separate carriageways crossing the water.

☒

The multiple images of the bridge are the result of several different pathways for the incident radar imagery to reflect back to the radar platform.

☒

Correct

That is correct

☒

The very high resolution evident on the main image of the bridge arch projected onto the water is higher than one would expect and is thus the result of operating the radar in the spotlight mode. Note that the (very faint) pillars seen joining the roadway to the arch are about 0.5m wide.

☒

Correct

That is correct.

☐

The very high resolution evident on the main image of the bridge arch projected onto the water is about what would be expected at X band. Note that the (very faint) pillars seen joining the roadway to the arch are about 0.5m wide.

